

13.4 Regularization

Weight decay, dropout, and related controls on capacity.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

Figures and notes follow Géron, *Hands-On Machine Learning*; Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*; and Theodoridis.

1. Why regularize

A deep net may have tens of thousands of parameters, sometimes millions. That freedom fits complicated data. It also **overfits**. You need **regularization**.

2. ℓ_1 and ℓ_2 (weight decay)

Both drop into backpropagation as a change to the loss. For ℓ_2 ,

$$J_{\text{reg}}(\theta) = J(\theta) + \lambda \sum_{i=1}^n \theta_i^2.$$

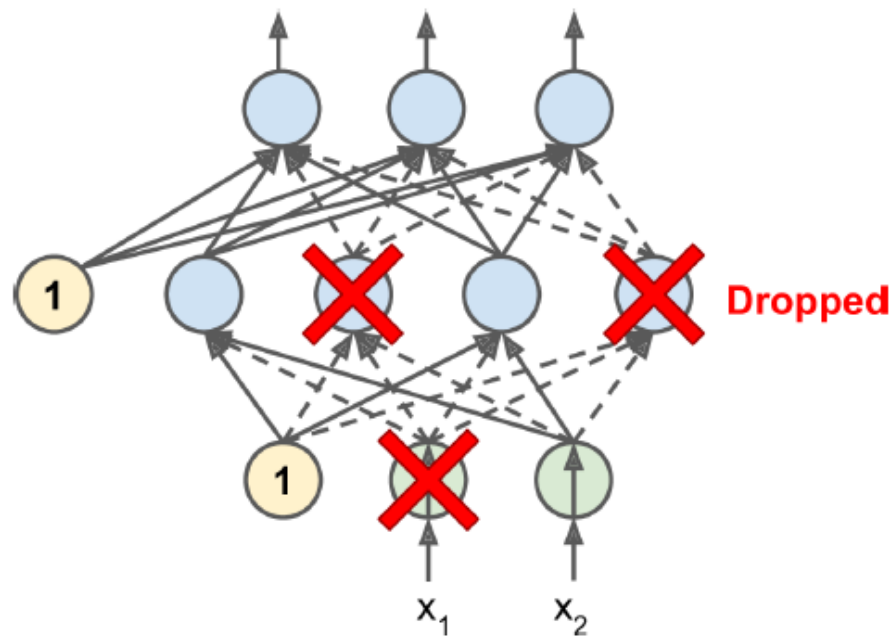
J is the original loss. λ is the strength.

On the backward pass you add the gradient of that penalty to the gradient of J . The update is ordinary descent on J_{reg} .

Effect: weights are **shrunk toward zero**. They cannot grow without bound. Larger λ means stronger shrinkage. This is **weight decay**: deterministic given θ .

3. Dropout

Hinton, 2012. At each training step, every neuron (inputs included, **outputs never**) is dropped with probability p : ignored for that step, maybe present on the next. p is the **dropout rate**, often **50%**. After training, nothing is dropped.



Overfitting: raise p . Underfitting: lower p . Large layers can take a higher rate than small ones. Many strong nets drop out only after the **last hidden layer** if full dropout is too strong.

Dropout **slows** convergence and usually **improves** the fitted model when p is right.

4. Monte Carlo dropout

Gal and Ghahramani, 2016.

- A net with dropout before every weight layer is **approximate Bayesian** inference. That is a justification, not a slogan.
- **MC dropout** can improve a *already-trained* dropout model: no retraining, no architecture change.
- It also gives a better **uncertainty** estimate.
- It is simple to implement.

Practice

1. Dropout versus weight decay: which one is stochastic at train time?
2. In ℓ_2 regularization, what does raising λ do to the weights?
3. You already trained a dropout net. What extra does MC dropout give you without a second training run?